NG JING YAO

Software Engineering Graduate | Full-Stack Software Developer | Applied AI

016-9568231 | jingyaong251104@gmail.com | https://www.linkedin.com/in/ng-jing-yao-89a172290/

PROFILE SUMMARY

Software Engineering graduate with experience in full-stack development and AI-powered applications. Experienced in Python, Flask, React, API integration, Docker, Retrieval-Augmented Generation (RAG), document retrieval, LLM APIs, LLM tool calling, and prompt engineering.

EXPERIENCE

Intern

YTL Construction | Jan 2026 - Jul 2026

- Developed and piloted a full-stack internal RAG chatbot for employees to retrieve IT and HR knowledge through a SharePoint-integrated chat interface.
- Built the frontend as a SharePoint Framework (SPFx) React/TypeScript extension with streaming responses, conversation memory, feedback collection, and error reporting.
- Collaborated with approximately 20 internal users to improve retrieval quality, knowledge coverage, answer reliability, and feedback workflows.
- Deployed separate chatbot services on Linux using Docker, with health-check endpoints, structured logging, feedback reporting, and local Windows development support.

Software Development Intern

YYC taxPOD | Nov 2023 - Feb 2024

- Assisted in developing web application modules using Vue.js and Laravel, contributing to user interface improvements and application functionality.
- Collaborated with the development team using GitHub for version control and front-end/back-end integration.

PROJECTS

Full-Stack RAG Chatbot

YTL Construction

- Built a full-stack RAG chatbot over 20+ internal IT and HR knowledge documents using Flask, LangChain, ILMU via OpenAI-compatible LLM APIs, and a hybrid retrieval pipeline with ChromaDB vector search, BM25 keyword search, and cross-encoder reranking.
- Created a document ingestion pipeline for Markdown/PDF knowledge sources, including hierarchical chunking, metadata tracking, abbreviation extraction, vector embedding generation, and incremental re-indexing.
- Implemented query preprocessing for conversation-aware query rewriting, language translation, abbreviation expansion, and multi-query generation to improve retrieval quality.
- Implemented 5 LLM tool-calling workflows with backend validation, structured tool results, and human-in-the-loop confirmation for sensitive actions.
- Designed guardrails for source-grounded answers, unsupported document claims, sensitive personal-data questions, and safe fallback responses.
- Configured separate containerized chatbot service profiles on Linux using Docker, with independent routing, environment variables, vector stores, and knowledge sources.

TECHNICAL SKILLS

Backend: Python, Flask, REST APIs, Java, PHP, Laravel
Frontend: React, TypeScript, JavaScript, Vue.js, HTML, CSS
Databases: MySQL, Firebase
Deployment: Docker, Linux, Internet Information Services (IIS)
AI/LLM: Retrieval-Augmented Generation (RAG), prompt engineering, LLM APIs, LLM tool/function calling, LangChain, ChromaDB, BM25, cross-encoder reranking
Tools: Git, GitHub, XAMPP, Codex, Claude Code

EDUCATION

Bachelor of Software Engineering

Tunku Abdul Rahman University of Management and Technology (TARUMT) | Jul 2024 - Aug 2026

CGPA: 3.73 | Dean's List: Nov 2024 | President's List: Jan 2026

Diploma in Information Technology

Tunku Abdul Rahman University of Management and Technology (TARUMT) | Jun 2022 - Jun 2024

CGPA: 3.79

LANGUAGES

English (Proficient) | Chinese (Native) | Malay (Intermediate)